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A gentle introduction to principal component analysis using tea-pots, dinosaurs, and pizza

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Abstract

Principal Component Analysis (PCA) is a powerful statistical technique for reducing the complexity of data and making patterns and relationships within the data more easily understandable. By using PCA, students can learn to identify the most important features of a data set, visualize relationships between variables, and make informed decisions based on the data. As such, PCA can be an effective tool to increase students data literacy by providing a visual and intuitive way to understand and work with data. This article outlines a teaching strategy to introduce and explain PCA using basic mathematics and statistics together with visual demonstrations.

KEYWORDS

correlation, covariance, data analysis, data literacy, data reduction, data visualization, teaching statistics, variance

1 | INTRODUCTION

The ability of extracting information from data in an effective way is one of the key components of data literacy.¹ Modern data are becoming increasingly complex in both quantity and dimensions: as a consequence, also data analysis has become more complex and more important in all disciplines, from economics to life sciences, from education to politics. It has been suggested that teaching data and statistical science should focus on tools that are suitable for dealing with modern data, rather than on tools designed for old data types.^{1,2} In view of the high dimensionality of modern data, it is also recommended that students are presented with data analysis tools capable of exploring and analyzing multivariate and high-dimensional data.^{1,2}

In this light, it is interesting to note that the most widely used tool for exploring multivariate and high-dimensional numeric data, Principal Component Analysis (PCA), was

introduced over 100 years ago! PCA was first introduced by Pearson in 1901,³ the same Pearson after whom the correlation coefficient is named. Later, Hotelling introduced a different formulation of PCA in 1933.⁴ Although the two formulations are equivalent, they are not identical in concept.^{5,6} PCA can also be expressed as a discrete Karhunen-Loève transform.⁷

PCA is widely used in virtually all disciplines where the structure of high-dimensional data needs to be explored, and the relationships between observations and the variables responsible for observed data patterns (the so-called *data structure*) need to be understood. Using PCA to explore, analyze and visualize data, students can develop a deeper understanding of data and the ability to work with it effectively.

Although a solid background in mathematics, including linear algebra, geometry, and calculus, is generally required to understand the theory of PCA, it may be necessary to introduce PCA to students in an applied context, focusing on

how and when to apply the method in the practical analysis of a data set or the extraction of relevant information from measured data. In such cases, a practical, computer-based approach may be favored while minimizing mathematical theory. However, to avoid the risk of PCA becoming another “black box” tool⁸ that produces results and figures with little or no context, students must have a certain understanding of the theoretical foundations of the method.

While PCA is usually introduced as a dimensionality reduction technique, we present here an approach to teach PCA starting from visualization, from which PCA as a dimensionality reduction technique follows at a later stage. PCA is introduced and explained using minimal mathematics, and it is presented as an effective tool to address the problem of visualizing high-dimensional data (in four or more dimension). The overall strategy is as follows:

1. Students are presented with the problem of visualizing a two-dimensional data set
2. Students are presented with the problem of visualizing a three-dimensional data set
3. Rotation in the plane and in the space is presented as a tool to explore and identify the data structure
4. Rotation is formulated using trigonometry, introducing the concepts of linear combination and weighted sum
5. A link is established between information content and variance
6. Rotation along the direction of maximum variance is presented which lead to the definition of:
7. Principal components and their meaning
8. The relationship between original variables, correlation and principal components is presented which leads to:
9. PCA as a data dimensionality reduction technique.

This approach has been developed (and used) for students at both the Bachelor and Master level in the Life Sciences programs at Wageningen University (the Netherlands), with class size ranging from 10 to more than 80 students. Classes were very composite, with students from different nationalities and backgrounds. This approach is, for most of the students, the first exposure to a multivariate tool for the analysis of high-dimensional data and informs following lectures and activities dedicated to more advanced tools (like sparse variant of PCA and PCA regression). The material is covered in two lecture classes (45 min each), each followed by hands-on computer practical classes (1.5 h each) in which students solves a set of exercises performing PCA using R and/or Matlab on provided data sets. Each practical class ends with a wrap-up session where results of the exercises are presented and discussed and feedback is provided.

This paper also presents the Principal Component Analysis of several data sets (some synthetic and some

for real life biology and biochemistry experiments) to provide first-hand experience on the interpretation of a PCA model. An appendix is included that presents a linear algebra formulation of the PCA problem and its solution.

We recently (October 2023) used this approach also in a course dedicated to first year graduate students with an experimental background in biological sciences and limited statistical knowledge who needed to develop basic data analysis skills: the proposed introduction to PCA was evaluated “Very good” or “Excellent” by all the students who filled in the evaluation questionnaire at the completion of the course. We indicate how we use this approach, and hope that others may find part or all of it useful for their teaching.

2 | VISUALIZING AND EXPLORING DATA IN TWO AND THREE DIMENSIONS

2.1 | Visualizing a data set in two dimensions

At the beginning of our lecture, students are presented with the following simple yet fundamental problem: We have a set of 4608 measurements of two variables, x and y , as shown in Figure 1A. The questions to be asked are:

1. How can we visualize this data and understand what it represents?
2. What is the structure of the data?

In the context of PCA, the “structure of the data” refers to the underlying patterns or relationships present in the data set.

A possible approach to explore the data structure is to use/attempt a spatial visualization, as every observation can be considered a point $P(x,y)$ on the plane, where each variable represents a spatial (orthogonal) coordinate Cartesian plane xOy (where xOy is a shorthand for a x - and y - axis reference system with axis origin in O). These points can be visualized by plotting each pair x and y , as shown in Figure 1B: the data set contains the x and y coordinates of a tea-pot! The tea-pot shape can be referred to as the structure of the data (set): the structure of the data set is a tea-pot.

2.2 | After rotation a tea-pot is still a tea-pot

Next we ask the students: “what if we rotate the tea-pot, say if we rotate it 65 degrees clockwise?” We obtain the

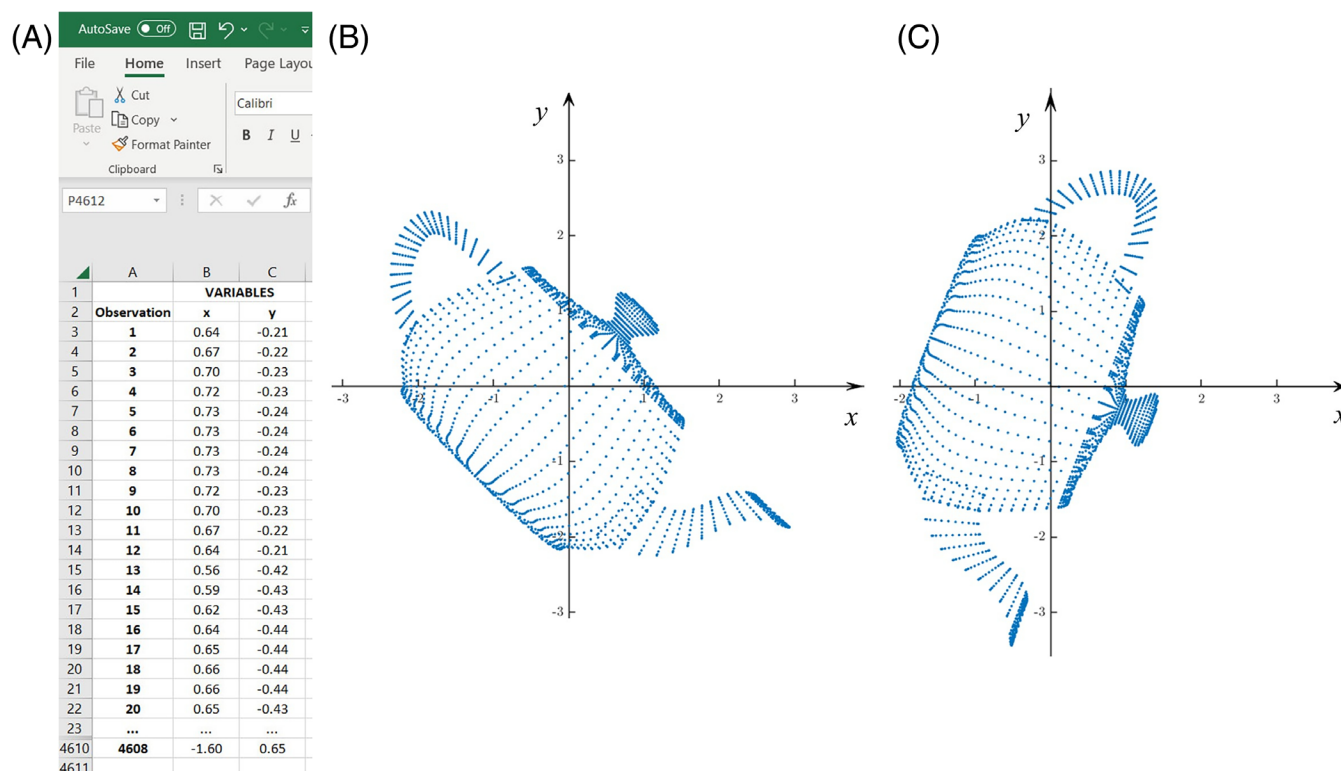


FIGURE 1 (A) Visualization of a two-dimensional data set $\mathbf{X}$. Screen-shot of a spreadsheet containing 4608 observations of two variables x and y which the students are asked to visualize. (B) Two-dimensional plot of the numerical data set $\mathbf{X}$ shown in panel A: It is a tea-pot! (C) You can recognize that it is a tea-pot even if it is rotated to be upside down. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/est.12363)]

right-hand the plot in Figure 1B. The answer of the students to this question should be immediate. Apart being now upside down, nothing really changed: we can still recognize 4608 points on the plane representing a tea-pot.

The rotation has not altered the structure of the data, nor our ability of recognizing the structure of the data: stated in other words, it has not made the structure of the data more (or less) clear to our eyes.

2.3 | Visualizing a data set in three dimensions

Extending the visualization approach to three dimensions is straightforward. When three variables are recorded, each observation can be represented in a three-dimensional space. However, this added dimension increases the complexity of understanding the data structure. We now consider a data set $\mathbf{X}$ where 36,876 observations are recorded for three variables x , y , and z , resulting in 36,876 points $P(x,y,z)$ that can be plotted in a three-dimensional space. This produces Figure 2A. The question we can ask students is, “What does it represent?” One student answered, “It is a dead cat run over by a

truck.” My response was, “Not exactly, but you are close—it is definitely a dead animal.”

In a three-dimensional space, we have more freedom to rotate the set of points. We can rotate around the x -axis, y -axis, or z -axis, or any combination. By rotating the points and using visual inspection taking a trial-and-error approach, we can arrive at Figure 2B, where the structure of the data becomes clear—it's a triceratops! The rotation allowed us to discover the true structure of the data set. This is why many statistical software programs provide three-dimensional plotting applications that allow plots to be rotated by the user. Rotations gave us a better perspective on the data set because since we can visualize the data point in three-dimensional space we could choose the best rotation, where “best” means a rotation that made us recognize the triceratops-like structure of the data.

This approach is limited, however, to three dimensions as we can only visualize objects in three dimensions. Also note that this visualization by trial-and-error can work only where the structure, or relationships among the variables, is quite clear (i.e., recognizable by our brain), and hence in a non-statistical (deterministic) situation. In statistics, we are seeking to detect structure or relationships in our data,

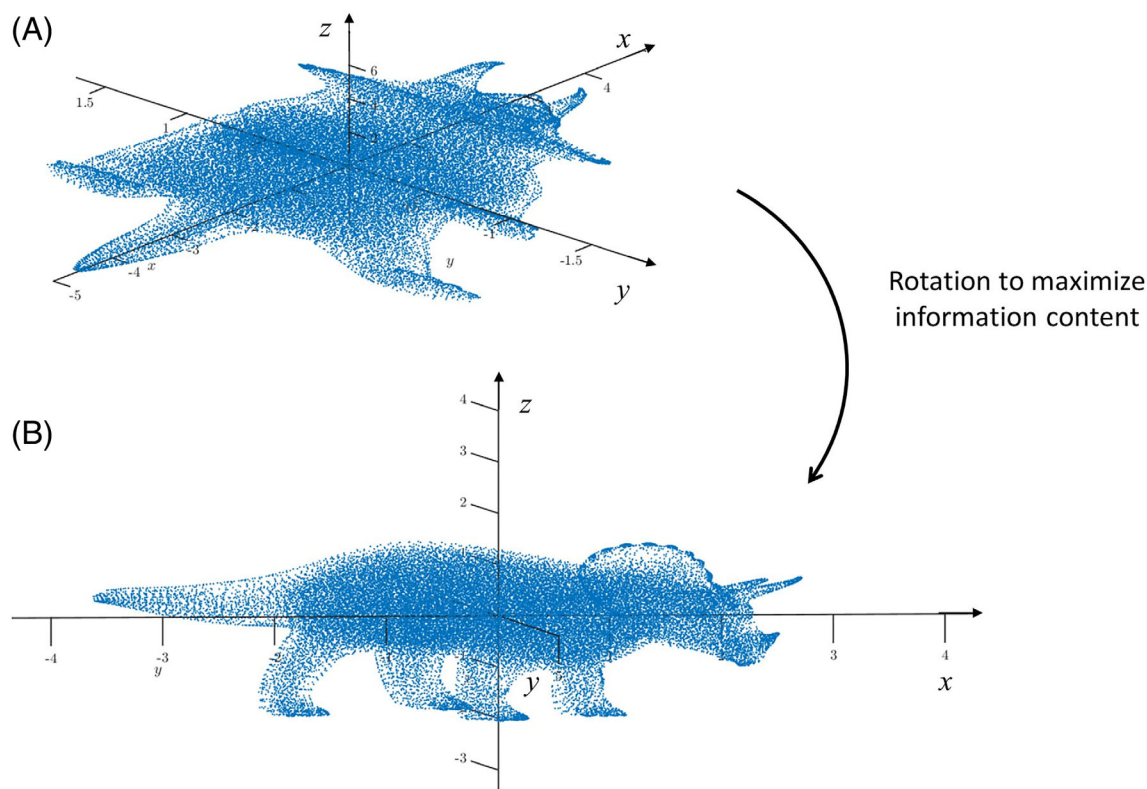


FIGURE 2 Rotation as a way to extract maximal information in the case of a 36876×3 data set $\mathbf{X}$. (A) Three-dimensional plot of the original data: there is structure in the data, but it is not clear what it is. (It is not a dead cat, as one student initially guessed). (B) Optimal rotation of the object: in this case the optimality is akin to a rotation that maximizes the possibility of recognizing the true structure of the data set. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/est.12363)]

allowing for variation around the structure which may not be recognizable but relevant for the problem at hand. We are also seeking to understand or explain the variation.

What if a data set has 4 variables or 1000 variables? It is not possible to visualize a 4-dimensional or 1000-dimensional object even if there is no random variation in the data. We can speculate about the existence of tea-pot or a triceratops in a p -dimensional space. In statistics, we are also trying to detect structure in the data while allowing for, and interpreting, variation: so, how can we find the structure of a high-dimensional data set? This is where Principal Component Analysis comes in, as a tool to detect and visualize the structure of high-dimensional data and obtain low-dimensional representation of complex, high-dimensional, data.

3 | VARIATION AS INFORMATION

The idea of information contained in a data set is fairly readily grasped by our students through the triceratops example: they can see that some directions in space reveal

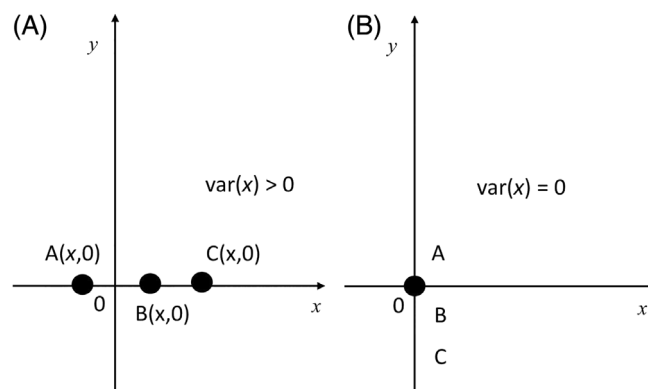


FIGURE 3 Variation as information. (A) Three points $A(x,y)$, $B(x,y)$, and $C(x,y)$ are scattered around $(0,0)$ and are distinguishable. The variance $\text{var}(x)$ of the x coordinate is larger than zero: the variation (differences) among the three points is here the information that allows to distinguish the three points: the information is quantified by the variance. (B) Three points $A(x,y)$, $B(x,y)$, and $C(x,y)$ are overlap on $(0,0)$ and are indistinguishable: since there is no variability among the coordinates there is also no information that allows to distinguish the three points.

the true form of the object (the structure of the data), while others obscure it. Thus, to identify the dinosaur, the directions of some rotations carry more information than others.

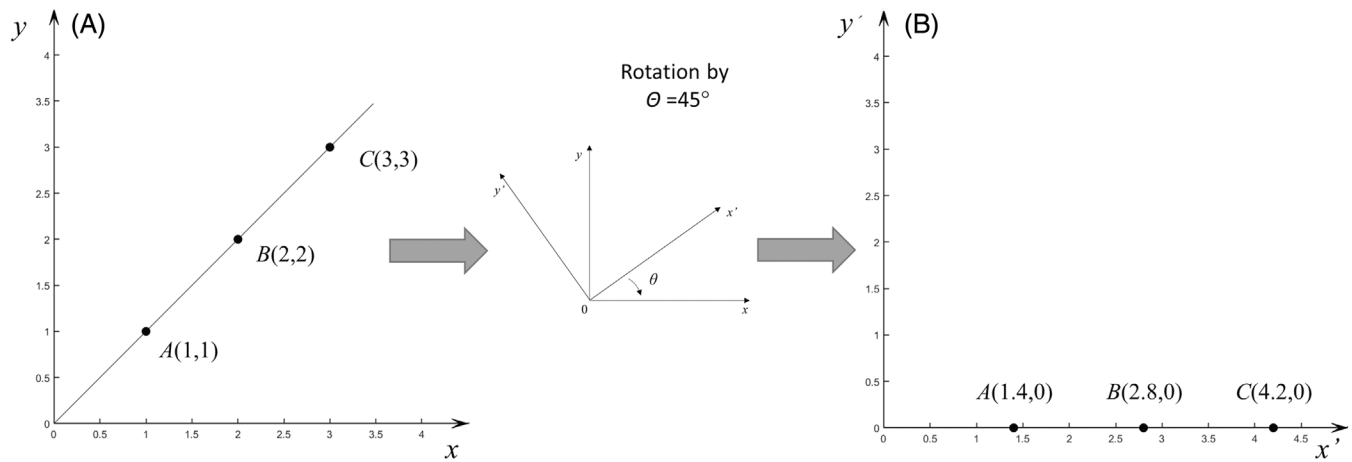


FIGURE 4 Rotation toward maximum variability. (A) Three points $A=A(1,1)$, $B=B(2,2)$, and $C=C(3,3)$ in the reference system xOy are aligned along the $\theta=45^\circ$ direction. The variance $\text{var}(x)$ and $\text{var}(y)$ of the x and y coordinates is 1. (B) The coordinates can be rotated by $\theta=45^\circ$ counterclockwise to define the new coordinate system $x'Oy'$ where the variance $\text{var}(x')$ of the new x' coordinate is 2 and $\text{var}(y')=0$. Rotation has maximized the variance along the x' direction. Note that the total variance is conserved: $\text{var}(x) + \text{var}(y) = \text{var}(x') + \text{var}(y') = 2$.

In this context, we now show how information can be understood as variation (variability): the more dispersed the data points are in a certain direction, the greater the information content along that direction. We first consider the simple example in Figure 3. In panel A, three points scattered around $x=0$ can be distinguished because their x coordinates (x values) are different. The diversity or dispersion of the three points is described by the variance. In Panel B, the three points perfectly overlap at $x=0$: there is no variation among the three points, making them indistinguishable and resulting in no information that can be used to determine how many points are present. Note that for the points in Panel A, a rotation a rotation of 90 degrees clockwise—or anti-clockwise—makes the three points indistinguishable.

4 | ROTATION ON THE PLANE

Mathematically, rotation on a plane or in the space can be introduced through the rotation matrix and the algebraic principles of linear transformations. However, for students with no knowledge of linear algebra but with a basic understanding of trigonometry, (i.e., the definition of sine and cosine by an angle θ), the rotation formula can be just presented. A point P of coordinates (x,y) is given in the reference system xOy . If the coordinate system is rotated of angle θ , the new coordinates of P in the rotated reference system $x'Oy'$ are given by the trigonometric formula:

$$\begin{aligned} x' &= x \cos \theta + y (-\sin \theta) \\ y' &= x \sin \theta + y \cos \theta. \end{aligned} \quad (1)$$

Equation (1) indicate how the new coordinates (x',y') are derived from the old coordinates (x,y) . The transformation is achieved through a linear combination of (x,y) , with coefficients determined by the sine and cosine of θ . These coefficients can also be interpreted as weights in a weighted sum, quantifying the importance of each old coordinate in the new coordinate system. This concept of weighting will play a crucial role in explaining the Principal Component Analysis (PCA) model, as will be discussed in Section 5.

5 | ROTATION TOWARD MAXIMUM VARIANCE DIRECTION

To demonstrate the concept of rotation toward direction of maximum variance, we consider the plot shown in Figure 4A showing the two-dimensional representation of a 3×2 data set $\mathbf{X}$ given by:

$$\mathbf{X} = \begin{bmatrix} x & y \\ A & 1 & 1 \\ B & 2 & 2 \\ C & 3 & 3 \end{bmatrix}. \quad (2)$$

In the xOy coordinate system, the points A , B , and C are located at $A(1,1)$, $B(2,2)$, and $C(3,3)$, respectively, and are aligned along the $\theta=45^\circ$ diagonal line. We focus now on the variance $\text{var}(x)$ and $\text{var}(y)$ of the x and y variables, that is we want to measure of their variability around their mean values. The mean values of the x and

y coordinates (variables) is 2, and the sample variance is, for both variables:

$$\frac{1}{2}((1-2)^2 + (2-2)^2 + (3-2)^2) = 1. \quad (3)$$

The variance along the x and y directions (i.e., the variance of the x and y coordinates), is equal to 1, implying that the amount of information along both axes is the same. Note that in this artificial example, we are treating the three points as sampled data in order to illustrate the concept. Because this is such an unsuitable sample to illustrate statistics, an instructor may wish to treat these three points as a population of three equally likely points, and use population variances; it makes no difference to the illustration of the concept.

It should be visually apparent that the reference system xOy can be rotated by 45° (counter clockwise), aligning all three points along the new x' axis.

The coordinates of point A in the new reference system $x'Oy'$ are now (This is also can be left to the students to calculate):

$$\begin{aligned} x' &= 1 \times \cos(-45^\circ) + 1 \times (-\sin(-45^\circ)) \\ &= \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = \sqrt{2} = 1.4 \\ y' &= 1 \times \sin(-45^\circ) + 1 \times \cos(45^\circ) = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = 0. \end{aligned} \quad (4)$$

For points B and C , the new coordinates (rounded to one decimal) are $(2.8, 0)$ and $(4.2, 0)$, as shown in Figure 4B. What about the variance of x' and y' ? Simple calculations give 2 and 0 for x' and y' , respectively. The variance in the x' coordinates of three points in the $x'Oy'$ reference system (Figure 4B) is now twice the variance of the x coordinates in the xOy reference system (Figure 4A), while the variance in the y' coordinates is now zero. Note that the total variance, that is, the sum of the variances in the two coordinate systems has not changed upon rotation and it is equal to 2.

The change of the variance in the coordinates occurred without modifying the relationships between data points: the distance $d(A, B)$ between the points A and B has not changed, as it can be readily verified by Pythagoras' theorem:

$$\begin{aligned} d(A(x, y), B(x, y)) &= \sqrt{(1^2 + (2-1)^2)} = \sqrt{2} = 1.4 \\ d(A(x', y'), B(x', y')) &= 2.8 - 1.4 = 1.4. \end{aligned} \quad (5)$$

The rotation has preserved distances. This we already had seen in the teapot example in Figure 1: rotating the

teapots has not affected the distances between the point, in fact after rotation, the teapot is still a teapot and the data structure has not been changed. However, the amount of information encoded along different dimensions is changed. In the $x'Oy'$ coordinate system, we see that the three points are indistinguishable along the y' direction and carry no information, as indicated by the variance of the y' coordinates being zero.

In contrast, the rotation by 45° degree has increased the variance in the x coordinate, which has gone from 1 (in xOy) to 2 (in $x'Oy'$). It is evident that the 45° line is the direction with the maximum spread of points in the xOy plane, and, therefore, it is the direction with the maximum variance in our data set.

We have found the direction of maximum variability for our 3×2 data set: the new coordinates x' and y' are called principal components of our data set.

The above example is artificial and not suitable to illustrate statistical data because the points lay exactly on a line in the xOy plane. We now consider a sample of three points as shown in Figure 5A: the direction of maximum variance will run along a diagonal line through the points, inclined at a certain angle θ with respect to the x -axis. How to determine the angle that maximizes the variance? For instance, consider a direction of 40° : after rotation, the sample variances for the transformed x' and y' (represented by the blue dots in Figure 5B) are, respectively, 4.7 and 1.3.

If we consider a rotation along the 57° direction, the sample variances in the new coordinates x' and y' (red dots in Figure 5B) are, respectively, 5.0 and 1.0. In xOy the variance along x is 2.3 or 38.3% of the total variance, since

$$\frac{2.3}{2.3 + 3.7} = 38.3\% \quad (6)$$

Equation (6) gives the so-called variance explained by direction x (see also Equation A11 in the Appendix A). If we rotate by 40° , the variance along x' is 4.7 or 78.3% of the total variance; rotating by 57° , the variance along x' is 6 or 83.3% of the total variance. As seen before, the total variance amounts to 6.0 in both coordinate systems.

The direction with an angle $\theta = 57^\circ$ is a better representation of the data set as it captures 83.3% of the total variance, while the direction with an angle $\theta = 40^\circ$ captures a bit less variance, 78.3% of the total variance. In PCA, this is referred to as "variance explained." The first principal component is defined as the direction that maximizes the variance, or the direction that explains the maximum variance. However, we cannot be certain that the 57° direction is the direction of maximum variance just relying on some level of graphical visualization and intuition.

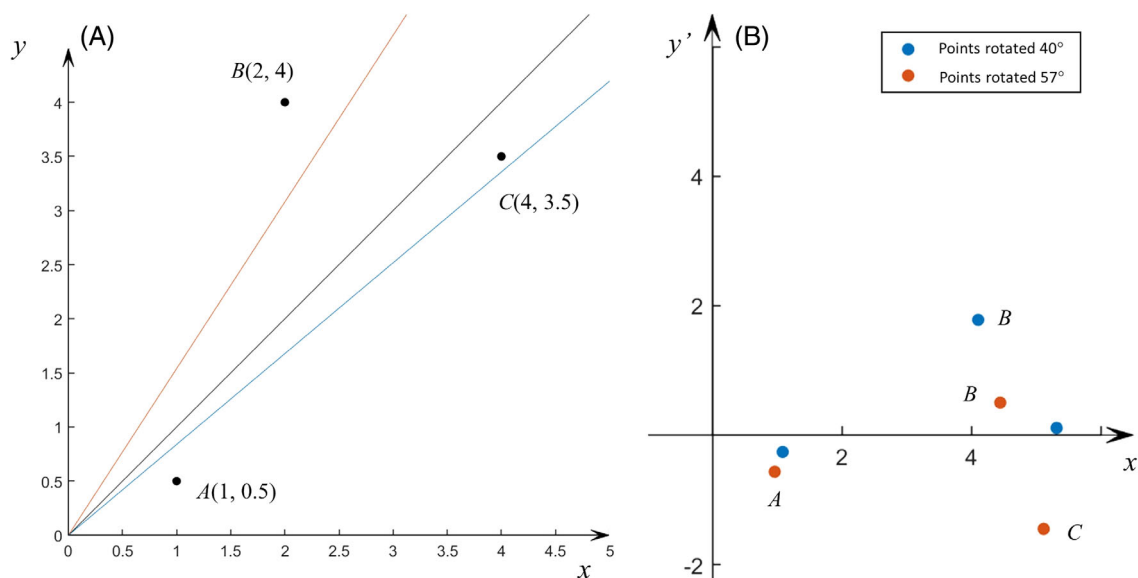


FIGURE 5 Three points scattered along the diagonal of the xOy plane. The variances of the x and y coordinates are 2.3 and 3.7, respectively (rounded to 1 decimal). (A) The direction of maximum variance is directed along an (unknown) angle θ . Two directions are shown for $\theta = 40^\circ$ (blue line) and $\theta = 57^\circ$ (red line). (B) The three points rotated in a $x'Oy'$ reference system, by $\theta = 40^\circ$ (blue points) and $\theta = 57^\circ$ (red points). [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/est.12363)]

Two key points should be emphasized to the students:

1. Determining the direction(s) of maximum variability becomes complex in high-dimensional data, as the number of angles required to determine the direction of maximum variance increases with dimensionality, that is, with the number of variables.
2. A trial-and-error approach is not efficient.

For this example the angle θ that maximize the variance along x is $\theta = 54.03^\circ$, for which the variance in x' is 5.0 (rounded): this solution is found using the mathematical method outlined in the Appendix A, which lies at the core of Principal Component Analysis and that can be easily implemented in statistical software. It is not necessary to delve into the technical details of how the principal components are actually calculated at this stage. Instead, the focus is be on illustrating and explaining the characteristics of Principal Component Analysis by considering the other very important result that can be derived from these two simple examples, namely that in each case, the principal components are uncorrelated. How instructors make this point depends on their students' backgrounds—whether they have previously seen correlation coefficients and whether instructors wish their students to obtain them here by hand or by using a calculator or dedicated software.

We can now state the general result for principal components of a numeric data set with n observations on p variables. The principal components are p new

variables which are linear combinations of the original variables such that:

1. the linear combinations of the data points are ordered by their sample variances. That is, the first principal component has the maximum variance, the second has the next highest variance, etc. and
2. the linear combinations of the data points have zero sample correlations, thus are uncorrelated.

Note that there are p principal components only when $n > p$. If $n \leq p$, there are $n - 1$ principal components. See the Appendix A for a linear algebra explanation.

6 | PRINCIPAL COMPONENT ANALYSIS AS DIMENSIONALITY REDUCTION

With reference to the three data points shown in Figure 4A, we observed that rotating along the $\theta = 45^\circ$ direction resulted in a new coordinate system, $x'Oy'$ (4B), where the y' coordinate of each point is zero and the points are perfectly aligned along the x' axis. In the $x'Oy'$ system, only one coordinate (x') is needed to determine the position of a data point, whereas in the original xOy system, two coordinates (x and y) were necessary.

The rotation along the 45° direction has reduced the dimensionality of the system from two to one. Why is this

possible? This happens because in this case there is a perfect linear relationship between the x and y coordinates. In fact $y=x$ which is a particular case of the more general linear relationship $y=ax+b$. The 45° rotation reduced the dimensionality from two to one because the x and y variables were perfectly correlated (Pearson's sample correlation $r=1$), as the points lay exactly on a line. In the context of PCA, this means that if two variables are perfectly correlated, one of them is redundant as they carry the same information. By rotating the plane, the two correlated variables x and y in xOy become two new uncorrelated variables x' and y' in $x'Oy'$. In fact, the two variables can be replaced by a new variable which is a weighted sum (i.e., a linear combination) of the two original variables, giving birth to a principal component.

This concept holds true also when there is no perfect correlation between observed/measured variables, as shown in Figure 6. The 25 points in xOy are spread around the $\theta=45^\circ$ direction and are highly correlated ($r=0.97$). The sample variances of x and y are both 1, thus the x and y direction accounts each for 50% of the total variance which is 2. The 45° rotation shown in Figure 6B reduces the original y coordinate and reduces the correlation between the two coordinates to $r=0.02$. The sample variances in the rotated system are 1.9 for x' and 0.1 for y' , whose direction accounts for 95% and 5% of the total variance (which is still 2).

The 45° direction is approximately the direction of maximum variability and the rotation along this direction maximizes the x coordinate while suppressing the y coordinate, effectively reducing the dimensionality. The

stronger the correlation between x and y , the more effective the suppression of the y variable. If x and y are perfectly correlated, the suppression of the y variable is perfect as there is variance only along one direction.

This example shows, in two dimensions, what PCA actually does: it identifies the direction(s) of maximum variability, converting correlated variables into uncorrelated variables, and this extends to any number of variables, that is, dimensions. As a result, the principal components are linear combinations of the original variables: for a two-dimensional case, the principal components, that is, the rotated variables, are given by Equation (1). The direction(s) of maximum variability, as well the coefficients for the linear combination for each variable, can be found as described in the Appendix A. In practice, we can use the principal components with the greatest variances to visualize and interpret the data. That is, we can choose to reduce the dimensionality of the data set. In the above example, we could reduce the dimensionality of the data from 2 to 1, given that the first principal component (the one given by x' summarize the 95% of the variance [or information] of the original data). Also, using the principal components can remove multicollinearity from the data.

7 | PRINCIPAL COMPONENT ANALYSIS OF A HIGH DIMENSIONAL DATA SET

We now proceed to analyze a high-dimensional data set $\mathbf{X}$ containing 36,876 observations of 1000 variables, thus

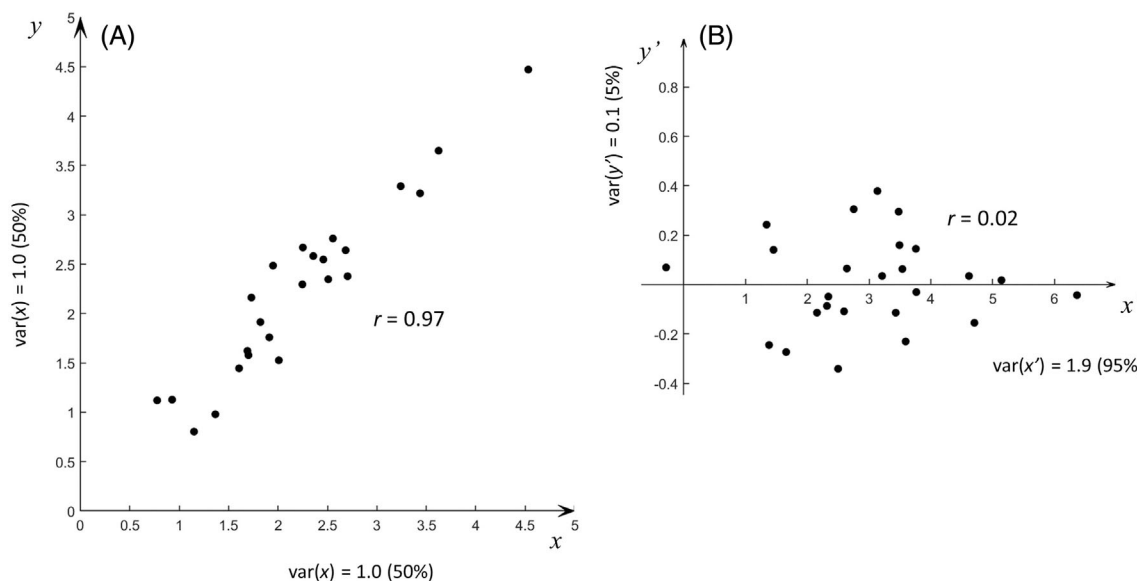


FIGURE 6 Rotation of correlated variables. (A) Two variables x and y that are correlated in the xOy plane: the sample correlation coefficient (Pearson's), is $r=0.97$. (B) The same variables rotated by 45° : the variables are now uncorrelated ($r=0.02$). Note that the scale y' -axis in panel B is larger than the scale of y -axis in panel A even if the value range is smaller: this is done to show how the rotated points are now scattered along the x' -axis in a random way, thus they are uncorrelated.

the data is 1000-dimensional: what is the structure of the data? We cannot visualize a 1000-dimensional data set and visualization is here not helpful. Before proceeding with the analysis, we give two new definitions:

1. The weights of the linear combinations defining the principal components are called *loadings*.
2. The representation of the original data in the principal component space (i.e., the rotated data in the PCA space) are called *scores*.

Let's now perform a PCA on the 36876×1000 data set **X** pictured in Figure 8A using Matlab. The command line is:

```
[R,T,~,~,VarExplained] = pca(X);
```

We obtain a 36876×1000 matrix **T** that contains the original 36,876 observations in a rotated coordinate system. There are still 1000 variables in **T**, so the rotated space is still 1000-dimensional. Each of the new coordinates in **T** is a linear combination of the original variables in **X**. The first principal component, that is the first new variable in **T** now accounts for most of the variance, followed by the second, and so on.

Figure 7B shows the variance explained for the first 5 principal components: the first two components account for around 50% of the total variance; the first three for

53%. The remaining variance is distributed over the remaining 997 dimensions. If we now plot the first two components, we obtain the two-dimensional representation of the 1000-dimensional data set which is given in Figure 7C.

The PCA has found rotations in the 1000-dimensional space such that 50% of the variance (information) present in the original data set is contained in the first two principal components. This means that if we only plot the first two component (which are, by definition, a linear combination of the original coordinates), we can represent 50% of the information contained in the 1000-dimensional data on the PC1-PC2 (bidimensional) plane (Figure 7C). By doing that (plotting 2 dimensions rather than 1000), we have reduced the dimensionality of the data set. As the figure shows, we have eliminated 998 variables (i.e., dimensions) but we retained sufficient information to describe the object (the triceratops), removing (i.e., denoising) dimensions that carry little information.

8 | INTERPRETATION OF PRINCIPAL COMPONENT ANALYSIS RESULTS

We now show the analysis and interpretation of the results of a Principal Component Analysis of a multivariate data set containing chemical data measured on 300 samples of

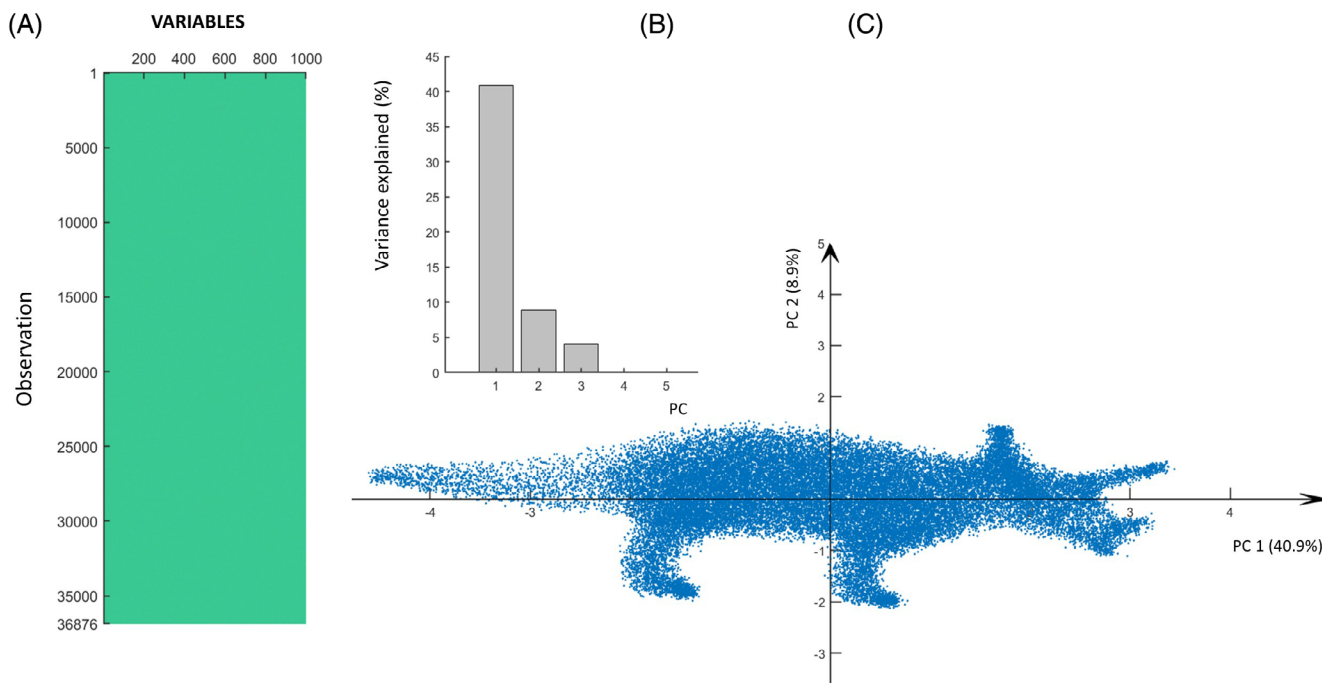


FIGURE 7 Principal Component Analysis of a high dimensional data set. (A) Screen-shot (color-coded) representation of a 36876×1000 data set **X**. (B) Variance explained by the first 5 principal components. (C) Plot of the first two principal components, giving a low-dimensional (2-dimensional) representation of the original 1000-dimensional data. The first two component accounts for 50% of the variance in the original data. This data set is derived from the three-dimensional Triceratops data set used in Figure 3 by expanding the original data set with 36,876 observations of 997 uncorrelated variables sampled by standard normal distribution. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/est.12363)]

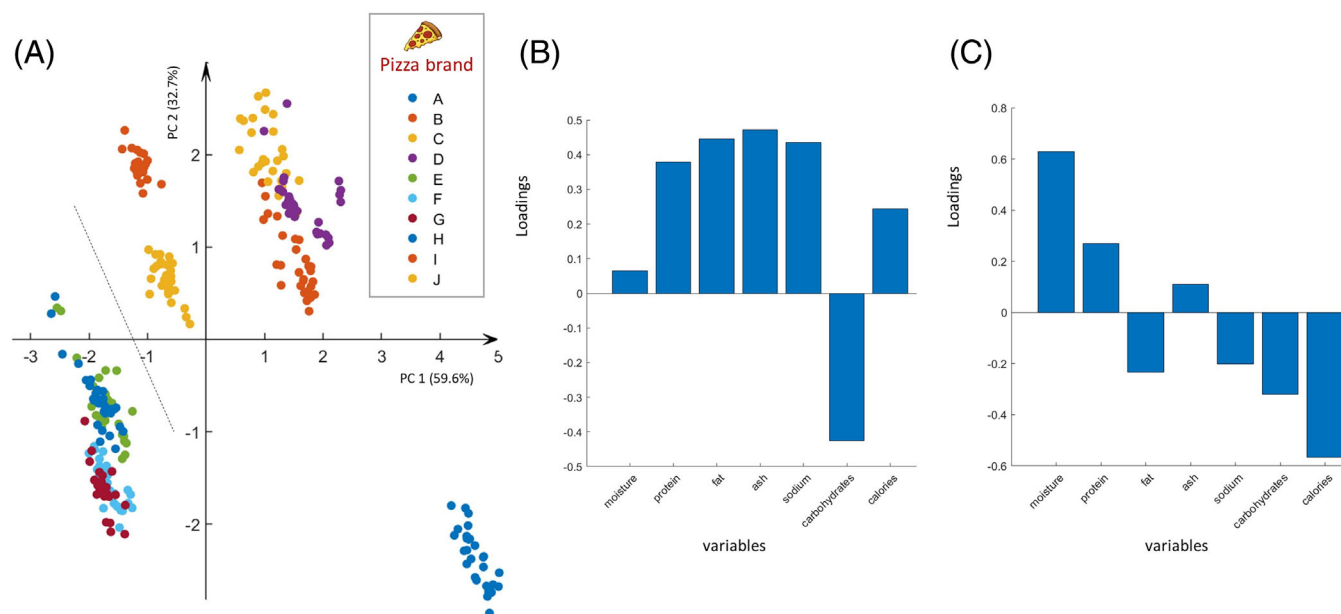


FIGURE 8 Principal Component Analysis of the pizza data set. (A) Score plot (i.e., the representation of the original seven-dimensional data in the principal component space) for first two principal components. (B) Loadings (variable importance) for the first principal component. (C) Loadings (variable importance) for the second principal component. The loadings represents the contribution of each variable to a given principal component, just as we have seen in the case of the two-dimensional examples in Figures 5 and 6. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/est.12363)]

pizza from 10 different commercial pizza brands (labeled A to J), on which seven chemical attributes have been measured: moisture (amount of water in the sample), protein, fat, ash, sodium and carbohydrates content and amount of calories (per 100 g of pizza). There are 30 samples for each brand, thus the data set $\mathbf{X}$ on which PCA is performed has size 300 (observations) $\times$ 7 (variables) and it is 7-dimensional.

Data are standardized to unit variance before analysis: see Section 10 Technical Remarks for more details about why this is (may be) necessary. The two-dimensional score plot (i.e., the representation of the original seven-dimensional data in the first two principal component space) is given in Figure 8A: each dot represents a pizza sample in the two-dimensional space defined by the first two principal component. Dots are color-coded according to the pizza-brand. The first two components accounts for more than the 90% of the total variance: the information that was originally contained in seven variables can be now described with just two variables (principal components). The pizza samples group in five larger groups, and each group containing pizza samples that share similar chemical profiles. Note that the information about to which brand each sample belongs is used only for visualization and it is not used in PCA: the clustering of the samples according to the brand (i.e., the data structure) is a characteristics of the data that PCA is able to condense in two dimensions. Pizza brand H appears to be

quite different from the other brands, since it separates from all other brands, as seen in the lower right quadrant of Figure 9A.

The plots of the loadings of the first and second principal components are shown in Figure 8B,C. The loadings, that are normalized such that the sum of their squared values equals 1 (see Appendix A for more details), give the importance of each original variable (in this case the chemical quantities) to a given component. The loading sign is arbitrary, because we can swap all the loading signs without changing the PCA model: what matters is the absolute value of the loading. In this example, we see that fat, sodium, ash and carbohydrates and calories are responsible for the difference between the pizza brands that separated along the first component, that is pizza brand that are different along the x-axis; moisture and calories content are important to differentiate among brands that separate along the second component, that is, along the y-axis. To separate the pizza clusters on the left side of Figure 8A, one needs to trace a diagonal line (indicated with a dashed line): this suggests that one or more variables contribute (i.e., have high loadings) to both components: in fact, calories have a high loading (in absolute value) on both components. The interpretation of a PCA model is context dependent, as well its relevance to the problem being studied: the importance of each variable should always be evaluated bearing in mind, in this case, its biochemical implications. This example was selected given that the student

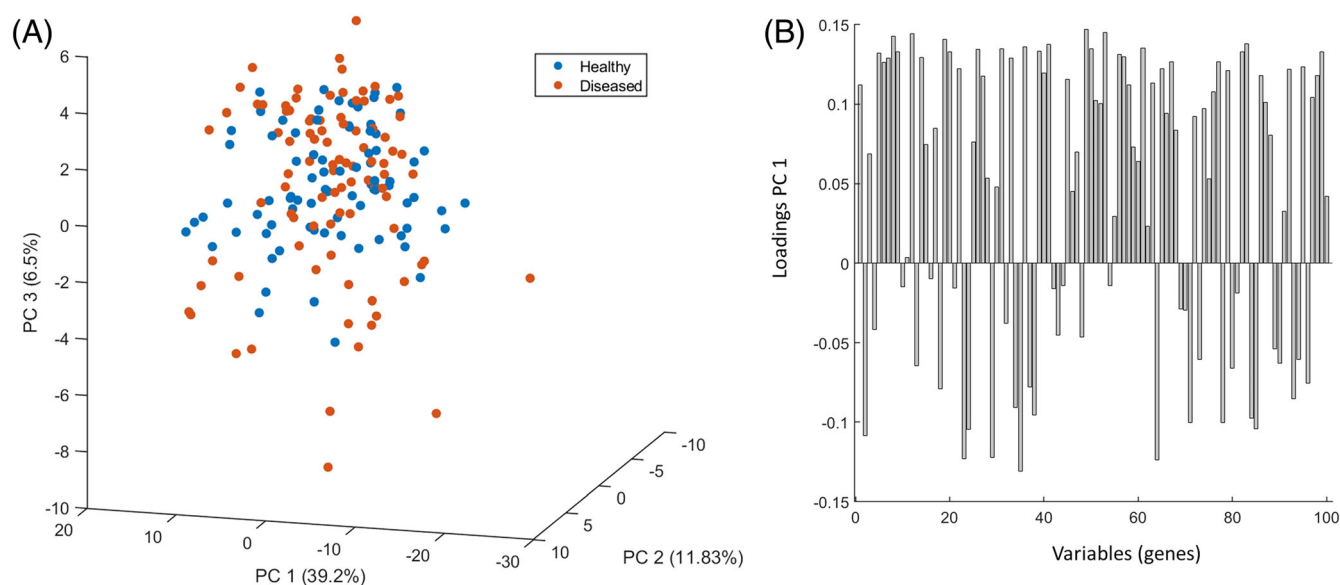


FIGURE 9 When PCA is not helpful. (A) Scatter plot of the two principal components of a PCA model of a data set containing 100 genes (variables) measured on the skin of 174 subjects, of which 92 (red dots) are suffering from a skin condition (psoriasis) and 82 are healthy (blue dots). (B) Loadings (variable importance) for the first principal component. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/west.12363)]

target group, for which this approach was initially developed, has a biological background. Other examples could be used to accommodate the needs of students from different disciplines.

9 | WHEN PRINCIPAL COMPONENT ANALYSIS DOES NOT SEEM TO BE HELPFUL?

A real life example where application of PCA to understand the structure of data does not result in a straightforward interpretation is shown in Figure 9. Here PCA is applied on a data set containing the gene expression values of 100 genes measured on the skin of 174 subjects, of which 92 are suffering from a skin condition and 82 are healthy. Gene expression data refers to the abundance of a mRNA transcript of a gene. The collection of the mRNA abundances of a set of genes measured on a biological sample is referred to as transcriptome or gene expression profile. For a primer on this topic see references 9, 10. The dimension of this data set is 174 (observations) $\times$ 100 (variables), thus the data set is 100-dimensional.

The interest is here into assessing whether the gene expression profiles of diseased subjects are different from those of healthy people. If so, one could expect the samples to separate in two (more or less) distinct groups (as it happens for some of the pizza samples in Figure 8). Figure 9A shows a three-dimensional plot of the first three components: dots are color-coded according to the

disease status disease (red dots) and healthy status (blue dots). We can see that there is no separation between the two groups. The variances explained by the three components are not tremendously different, suggesting that in the data, there not privileged directions along which variance (information) is concentrated. This happens because the original variables are poorly correlated and we have discussed how PCA exploits the correlation among variables: in fact if the variables are uncorrelated to start with, PCA will return the original variables but ordered by their variances. Basically, if the variables are uncorrelated, there are no directions of maximum variance: reference 11 has nice visual examples of this situation and discusses the problem and its consequences for PCA in gentle but rigorous way.

This can be seen in this example by the fact that the data points in the three-dimensional PCA space are roughly shaped like a ball. Examining the loadings (as we have done for the pizza example in Figure 8) highlights another limitation of PCA. The loading plot for the first principal component (similar plots are obtained for the other components) shows how all variables contribute almost equally to the definition of the principal component. This also illustrates an inherent disadvantage of PCA for some data sets: the new variables (i.e., the principal components) are linear combinations of all the original variable: as such they are not necessarily as easy to interpret as the original variables in themselves. Their main use is to identify high contributions to variation and to be able to consider fewer variables—that is,

reduced dimensionality—in visualization and/or analysis. Variants exist of PCA that aim to provide simpler models by suppressing the contribution of sets of variables: this is the realm of the so called Sparse-PCA, but this is a rather advanced topic. The interested reader can refer to references 12–14 for an overview.

For this example, and based on the PCA analysis, we can conclude that there is no difference, in the three-dimensional space, among the gene expression profiles of diseased and healthy people. It may happen that a difference between the two groups exists in a higher-dimensional space. For this data set, PCA is not helpful or, better stated, this data set has a structure that cannot be exploited using PCA.

To summarize, PCA is useful for data exploration and dimensionality reduction only when there are correlations among the variables. We emphasize again that PCA produces linear combinations that are uncorrelated and ordered by their (sample) variances from maximum to minimum. If the original variables are uncorrelated, or nearly so, PCA gives back the original variables (or close to). Also correlation (at least the one employed in PCA, which is Pearson's correlation) is only a measure of linear relationship, so curved relationships will be hidden and could even inhibit data understanding and interpretation.

10 | SOME TECHNICAL REMARKS

In real life applications, (measured) variables may have significantly different ranges, units, or variances. When the variables have different scales or units, those with larger magnitudes can dominate the analysis, leading to biased results, in the sense that the principal components will be dominated by the (few) variables with the larger variance and/or magnitude. To avoid this, variables are often scaled to a common scale, for instance by subtracting the mean and dividing by the standard deviation: in this way, all variables have zero mean and variance equal to 1. This procedure is called standardization, or auto-scaling, or scaling to unit variance (the “standardization” tends to be preferred in statistics as data science tends to differentiate between “scaled” [to a given common scale] and “standardized” by mean and standard deviation for each variable.) PCA is extremely sensitive to scaling or data transformations, and interpretation also depends on how the data have been processed. Reference 15 is an excellent introduction to this problem.

PCA, as described here, is applied on numerical data expressed on a ratio scale. Variants of PCA have been proposed to analyze different types of data, like binary data (i.e., data consisting of 0 and 1): see, for instance, references 16–18 and references therein.

Another crucial step in PCA is the determination of the optimal number of principal components. The problem arises from the need to strike a balance between preserving enough information and reducing the dimensionality of the data. Selecting too few components may result in a loss of information, while choosing too many components can lead to overfitting and increased computational complexity. Various methods addressing this problem have been developed to address this problem including scree plots, cumulative explained variance, cross-validation techniques and statistical methods. These approaches aim to identify the components where the additional explained variance becomes negligible, allowing for the selection of an optimal number of principal components that effectively represent the data. The literature on the topic is huge, but the mathematics needed is, in most case, quite advanced. The problem is introduced in a discursive way in reference 5; reference 19 provides a higher level introduction to the problem and to dimensionality assessment approaches.

11 | CONCLUSIONS

We have provided a gentle introduction to Principal Component Analysis that uses a limited amount of trigonometry and basic statistical concepts such as variance, covariance, and correlation. This approach is introduced by visual intuition of rotations in the plane and space, making it suitable for introducing PCA to high-school students or first-year university students in programs that require an introduction to data analysis techniques without assuming familiarity with linear algebra.

12 | SOFTWARE AND DATA

The tea-pot coordinates (Figures 1 and 2) have been obtained from Matlab using the built-in command: `[coordinates, x, y] = teapotGeometry`. The triceratops data (Figure 5) has been obtained from https://gitlab.com/e62Lab/medusa/-/blob/dev/examples/poisson_equation/triceratops_domain.h5. The 3-D coordinates have been extracted to Matlab format using the command: `coordinates = h5read('triceratops_domain.h5','/domain/pos')`. The pizza data set is available at: <https://data.world/sdhilip/pizza-datasets>.

The psoriasis data (a selected subset of the data published in²⁰) it is available in the gitlab repository associated to this article.

Data and Matlab codes are available <https://gitlab.com/esaccenti/dinosaurpca>.

All analysis was performed in MATLAB version: 9.14.0 (R2023a), Natick, Massachusetts: The MathWorks Inc.; 2022.

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CONFLICT OF INTEREST STATEMENT

The author declares no conflict of interest.

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APPENDIX A

In this Appendix, a linear algebra formulation of the PCA solution is given. We organize n observations of p variables in a $n \times p$ matrix $\mathbf{X}$. The variables are the columns of $\mathbf{X}$: $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p$. Each variable is thus a $n \times 1$ vector. We assume each column $\mathbf{X}$ in to be mean centered.

A linear combination $\mathbf{t}$ of the p variables (columns of $\mathbf{X}$) is $n \times 1$ vector given by

$$\mathbf{t} = w_1 \times \mathbf{x}_1 + w_2 \times \mathbf{x}_2 + \dots w_p \times \mathbf{x}_p, \quad (\text{A1})$$

where $w_1, w_2, \dots, w_p$ are numerical coefficients (weights). We have stated that principal components are linear combinations of the original variables that maximizes variance. Thus the problem is to find the appropriate set of weights that maximize the variance of $\mathbf{t}$.

In the linear algebra formulation, the relationship between $\mathbf{t}$, $\mathbf{X}$ and the weights w can be written as

$$\mathbf{t} = \mathbf{X}\mathbf{w} \quad (\text{A2})$$

where now $\mathbf{w}$ is a $p \times 1$ vector whose elements are the $w_1, w_2, \dots, w_p$ weights.

Since the columns of $\mathbf{X}$ are centered, also $\mathbf{t}$ is centered, that is, its mean value $\bar{\mathbf{t}}$ is zero. The variance $\text{var}(\mathbf{t})$ of $\mathbf{t}$ is given by

$$\begin{aligned} \text{var}(\mathbf{t}) &= \sum_{i=1}^n (t_i - \bar{\mathbf{t}})^2 = \\ &= (t_1)^2 + (t_2)^2 + \dots + (t_n)^2 \\ &= \mathbf{t}^T \mathbf{t}. \end{aligned} \quad (\text{A3})$$

From Equations (A1) and (A3), it descends that since $\mathbf{t}$ depends on $\mathbf{w}$, also the variance of $\mathbf{t}$ depends on the weights. It should be noted that once the optimal weights maximizing $\mathbf{t}$ have been found, we can arbitrarily increase the variance by just multiplying $\mathbf{w}$ by any (large) number, this because of the general property of the variance $\text{var}(a\mathbf{z}) = a^2 \text{var}(\mathbf{z})$. Thus, to have a properly specified problem with a unique solution, we require the weights $\mathbf{w}$ to be normalized, that is we require $\|\mathbf{w}\| = 1$, that is $\mathbf{w}^T \mathbf{w} = w_1^2 + w_2^2 + \dots + w_p^2 = 1$. Note that the rotation coefficients in Equation (1) satisfy this condition since $\sin^2 \theta + \cos^2 \theta = 1$. This is true for any orthogonal rotation in a p -dimensional space.

Considering Equations (A2) and (A3), we can write:

$$\text{var}(\mathbf{t}) = \mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w}. \quad (\text{A4})$$

Assume now that $\mathbf{w}$ is the set of weights that give maximum variance λ , so that

$$\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} = \lambda. \quad (\text{A5})$$

Since we impose that $\mathbf{w}^T \mathbf{w} = 1$, we can write:

$$\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} = \lambda \times 1 = \lambda \mathbf{w}^T \mathbf{w}, \quad (\text{A6})$$

from which:

$$\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - \lambda \mathbf{w}^T \mathbf{w} = 0, \quad (\text{A7})$$

and:

$$\mathbf{w}^T (\mathbf{X}^T \mathbf{X} \mathbf{w} - \lambda \mathbf{w}) = 0. \quad (\text{A8})$$

Since $\mathbf{w}$ is non-zero, Equation (A8) is satisfied only and only if

$$\mathbf{X}^T \mathbf{X} \mathbf{w} = \lambda \mathbf{w}. \quad (\text{A9})$$

Equation (A9) is the standard eigenvalue-convector problem: this relationship is satisfied only when the vector $\mathbf{w}$ is an eigenvector of the matrix $\mathbf{X}^T \mathbf{X}$, and λ is the associated eigenvalue. The $p \times p$ matrix $\mathbf{C} = \mathbf{X}^T \mathbf{X}$ is the sample variance-covariance matrix obtained from the data matrix $\mathbf{X}$. It follows that:

- The vector(s) $\mathbf{w}$ that maximizes the variance is (are) given by the eigenvectors of the covariance matrix. For instance, the first eigenvector of $\mathbf{C}$ gives the coefficient of the rotation of $\mathbf{X}$ that gives the linear combination of the original variables defining the first principal component.
- The variance of each principal component is given by the eigenvalue associated to the corresponding eigenvector.

Using a more precise notation::

$$\mathbf{C} = \mathbf{X}^T \mathbf{X} = \mathbf{U} \mathbf{L} \mathbf{U}^T = \sum_{j=1}^J \lambda_j \mathbf{u}_j \mathbf{u}_j^T \quad (\text{A10})$$

where $\lambda_1, \lambda_2, \dots, \lambda_p$ are the p eigenvalues of $\mathbf{C}$ arranged in descending order on the $p \times p$ diagonal matrix $\mathbf{L}$. The orthonormal eigenvectors $\mathbf{u}$ (i.e., principal components) collected as columns of $\mathbf{U}$. The eigenvalue λ_j measures the variance of the corresponding j -th principal component, and the eigenvectors $\mathbf{u}_j$ is the vector of the

loadings, that is, the coefficients needed to define the uncorrelated j -th principal components. The fraction of variance v_j explained by the j -th component is given by

$$v_j = \frac{\lambda_j}{\sum_j \lambda_j}. \quad (\text{A11})$$

Operatively, given $\mathbf{X}$ (centered), the principal components are found by eigendecomposition of the covariance matrix.

Note that if the number of samples n is larger than or equal to the number of variables p ($n \geq p$) and the matrix $\mathbf{X}$ is full rank (i.e., $\mathbf{X}$ has p linearly independent columns, $p - 1$ if $\mathbf{X}$ is centered as customary in PCA), then the covariance-correlation matrix $\mathbf{C}$ has rank p , and has p distinct eigenvalues. If the number of samples n is smaller than the number of variables p ($n < p$) $\mathbf{X}$, then $\mathbf{C}$ has maximum $n - 1$ eigenvalues, and thus there are maximum $n - 1$ principal components. What happens if the eigenvalues of $\mathbf{C}$ are not distinct is discussed.¹¹